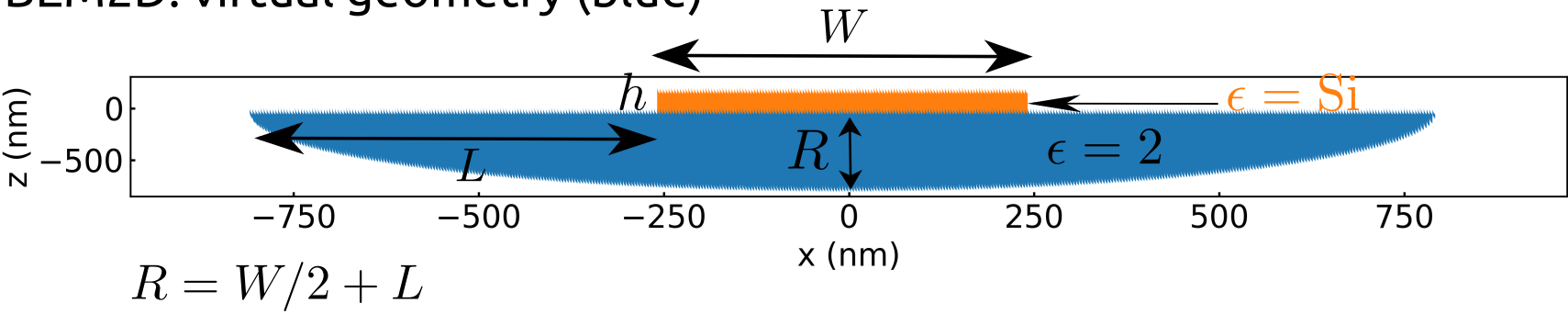


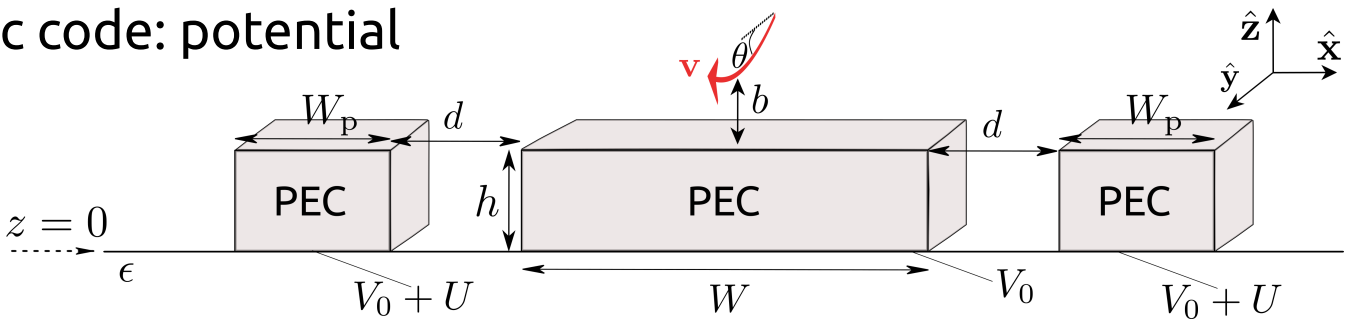
Electron-coupling to waveguide: EELS

$E_e = 200 \text{ keV}$

BEM2D: virtual geometry (blue)

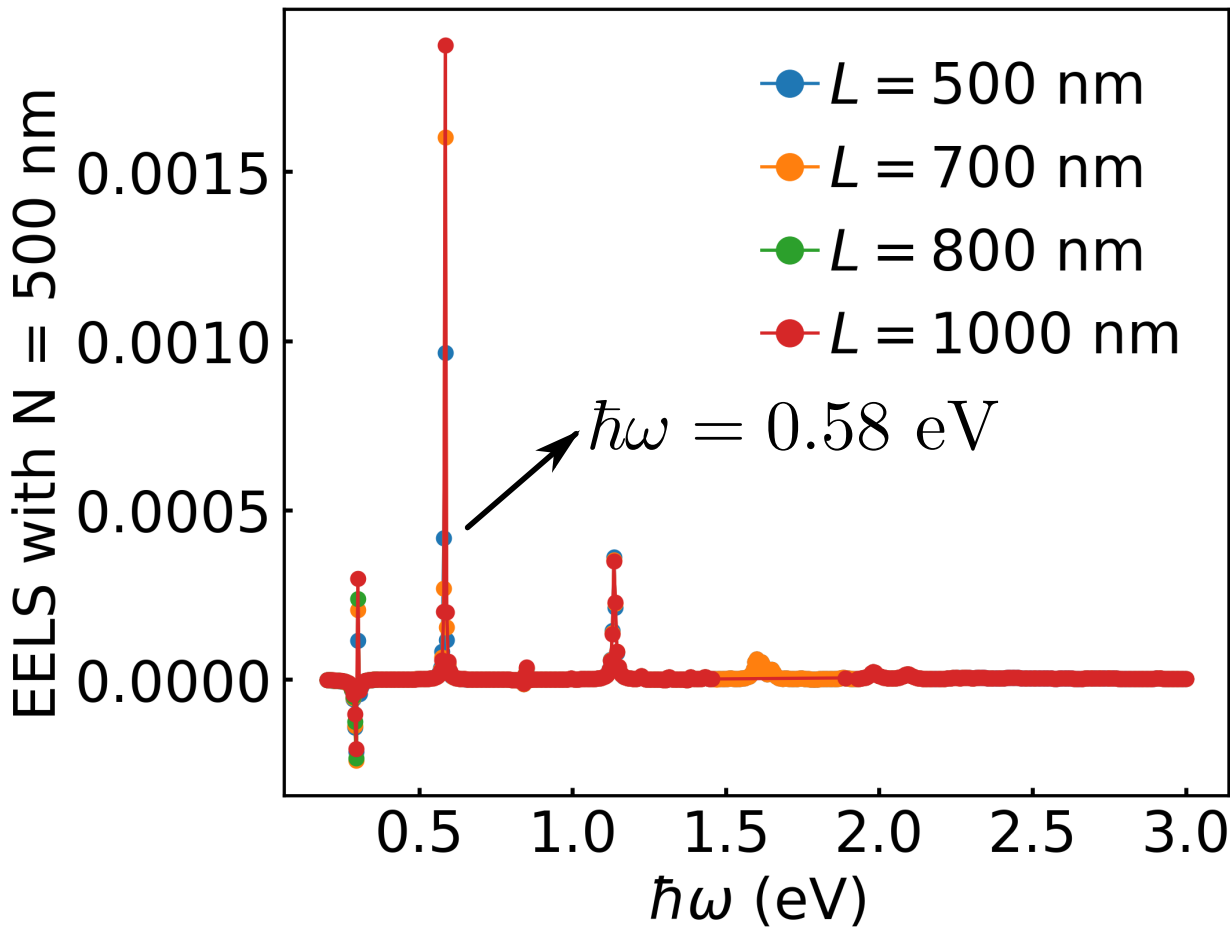


c code: potential



Running EELS for different L :

$W = 500 \text{ nm}$ $h = 300 \text{ nm}$



height of the peak
depends on L

Writting the integration of EELS in c to run in sonic: ✓

$$\Gamma_{\text{EELS}}(\omega) = 2 \int_{b_{\min}+h}^{\infty} \frac{vdz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(x,z)}{m_e \gamma_e} \right|}} \frac{d\Gamma_{\text{EELS}}(\omega, x, z)}{dy}$$

```
Data saved for j = 115, V0 = 0.136 eV, theta_mrad = 0.826
Data saved for j = 116, V0 = 0.136 eV, theta_mrad = 0.827
Data saved for j = 117, V0 = 0.138 eV, theta_mrad = 0.834
Data saved for j = 118, V0 = 0.139 eV, theta_mrad = 0.836
Data saved for j = 119, V0 = 0.141 eV, theta_mrad = 0.842
```